

RESEARCH INTERESTS

Robotics | Safety-Critical Control Systems | Autonomous Systems | Motion Planning & Navigation

EDUCATION

Degree	Institute	CGPA	Year of Graduation
Master of Science	University of Michigan, Ann Arbor	4.0/4.0 (current)	2027 (expected)
Bachelor of Technology	Indian Institute of Technology, Kharagpur	8.77/10	2025

TECHNICAL SKILLS

- **Languages:** Python | MATLAB | C | Arduino | Visual Basic | LaTeX
- **Software:** ROS | Gazebo | RViz | MATLAB/Simulink | Fusion 360 | Pix4Dmapper

EXPERIENCE

- **Graduate Research | Distributed Autonomous Systems and Control Lab | Prof. Dimitra Panagou** *September 2025 - Present*
 - Implementing and analyzing safety filters: **backup control barrier functions**, **model predictive shielding**, and the **gatekeeper**
 - Investigating the core theoretical difference between these three safety filters on pre-defined backup and nominal controllers
 - Designed and simulated a series of scenarios to demonstrate the theoretical differences and the failure modes of each filter
- **Undergraduate Research | Unmanned Aerial Vehicles Lab | Prof. Sikha Hota** *August 2024 - July 2025*
 - Implemented **Biased PNG**, **Two-Stage PNG**, and **Time-to-go Polynomial Guidance** laws for 2D missile interception scenarios
 - Validated that the guidance algorithms achieved **desired impact angles** while satisfying **acceleration and FOV constraints**
 - Adapted **multi-planar** guidance laws for **3D interception** and worked to eliminate lateral acceleration profile discontinuities
 - Designed a **2-planar 3-Staged 3D** Impact Angle Control Guidance (**IACG**) law that can achieve any terminal impact direction
- **Research Intern | RWR&DC | Hindustan Aeronautics Ltd** *May 2024 - July 2024*
 - Analyzed, in-depth, the **AFCS** (Automatic Flight Control System) and actuator mechanisms of a semi-autonomous helicopter
 - Integrated **Pitt-Peters** dynamic inflow model into the helicopter dynamics simulator to improve transient response predictions
 - Validated simulation results against flight test data, for **collective dump** input, identifying areas for future developments
- **Summer Intern | Aircraft Division | Hindustan Aeronautics Ltd** *June 2023*
 - Monitored manufacturing lifecycles ranging from machining to final assembly for **Hawk**, **Jaguar**, and **LCA Tejas** aircrafts
 - Evaluated system integration protocols through immersive technical rotations in aircraft hangars and production facilities

PROJECTS

- **Aerial 2D Mapping and Estimation of Forest Area Cover | Prof. Sandeep Saha** *Jan 2024 - Apr 2024*
 - Automated forest cover estimation using a **U-Net CNN** to support environmental monitoring and archaeological surveys
 - Constructed orthorectified 2D maps from aerial images, having GPS and altitude data, by stitching using **Pix4Dmapper**
 - Enhanced the map resolution using **EDSR** and estimated average forest canopy height for sample data by creating a **DSM**
- **Design and Analysis of a Quad-Wing Flapping Prototype | Prof. Sandeep Saha** *Aug 2023 - Nov 2023*
 - Designed and built a **dragonfly-inspired** quad-wing flapping prototype with a **gear-slider** mechanism for **variable wing phase**
 - Integrated a **passive joint** to allow wing rotation for realistic lift generation and conducted smoke visualization experiments
 - Quantified force changes with varied wing beat frequencies and front-wing-hind-wing phase differences, using **precision sensor**
 - Analyzed spanwise and chordwise wing flexibility effects on lift using **3** different wing materials, utilizing a high-speed camera
- **Aerial Carbon Emission Monitoring System | Prof. Srinibas Karmakar** *Aug 2023 - Nov 2023*
 - Designed a drone-deployed module equipped with **MQ7** and **MQ135** sensors and measured **CO** and **CO2** emissions real-time
 - Developed a method for ascent time calculations, intended for a **helium-balloon-lifted setup**, adopting an atmospheric model
 - Engineered an altitude tracking system using **BMP280** sensor data and employed **nRF24L01** modules for data transmission
- **Smart Irrigation System | Prof. Manjunatha Mahadevappa** *Feb 2022 - Mar 2022*
 - Implemented an automated irrigation system, using **Arduino**, optimizing water usage based on **soil moisture** and **temperature**
 - Integrated and calibrated moisture sensor, **LM35** and a water pump, and developed embedded software to control irrigation

PUBLICATIONS

1. N. K. Singh, S. Pal, **A. D. Menon**, and S. Hota, “Impact Angle Constrained Guidance Laws for Stationary Target Interception in 3D Space,” in *AIAA SCITECH 2026 Forum*, 2026, p. 2348.

RELEVANT COURSEWORK

Online Learning for Control | Flight Vehicle Controls | Flight Stability & Control | Automatic Control of Aircraft
Navigation and Guidance of Aerospace Vehicles | Space Dynamics | Mechanics of Flight | Intermediate Dynamics
Motion Planning and Control | Embedded Control Systems | Systems Laboratory | Helicopter Engineering | Robotics